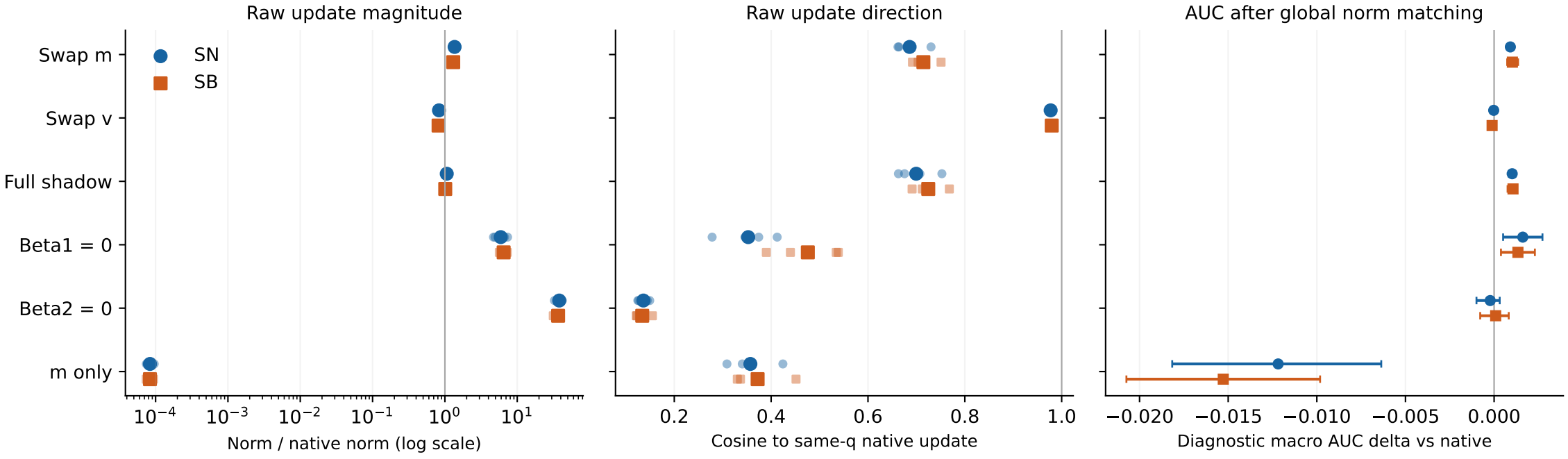
V11 local state interventions at epoch 10: four training seeds
Raw update magnitude
Raw update direction
AUC after global norm matching
Swap m
Swap v
Full shadow
Beta1 = 0
Beta2 = 0
m only
SN
SB
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
Norm / native norm (log scale)
0.2
0.4
0.6
0.8
1.0
Cosine to same-q native update
−0.020
−0.015
−0.010
−0.005
0.000
Diagnostic macro AUC delta vs native
Geometry: dots = seeds; solid markers = means. AUC bars: existing seed-bootstrap 95% intervals.
Native-SN anchors, same current q and batch. Global norm matching does not fix block norms or establish long-run mediation.